Rotoforge Studio — User Manual

For the Rotoforge AFRB (additive friction rotational bonding) wire-deposition machine. Manual snapshot 2026-07-03; screenshots are captures of the real application slicing a real part against a real FRAM process-window export.

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1. What Rotoforge Studio is

Rotoforge Studio turns a 3D mesh into RepRapFirmware G-code for the Rotoforge AFRB machine: X, Y, Z motion, the rotary wheel axis `A`, and the wire feeder `E`. It is not a filament slicer — deposition happens by friction between a spinning wheel and fed wire, which imposes hard rules a normal slicer does not have:

- **No grinding** — a wheel in contact must always be moving fast enough AND feeding wire; anything else removes material instead of adding it.
- **All dwells airborne** — the spindle settles and stabilizes only while lifted.
- **Tangential tool** — the wheel always faces its direction of travel; curves are limited by how fast the axis can slew and by its usable angular range.
- **The wire never retracts** — pass separation is a mechanical cut.
- **Constant revs/mm within a pass** — RPM and traverse speed stay fixed for the whole pass and only change airborne, between passes.

The part you should trust: **the G-code emitter re-proves every one of these rules on the final output.** If any check fails, the studio shows a validation error and refuses to produce G-code. A red "VALIDATION FAILED" message is the safety system working, not the program breaking.

The workflow:

> **Import mesh → Load process window (material) → Pick settings → Slice → Inspect preview / simulation → Save G-code**

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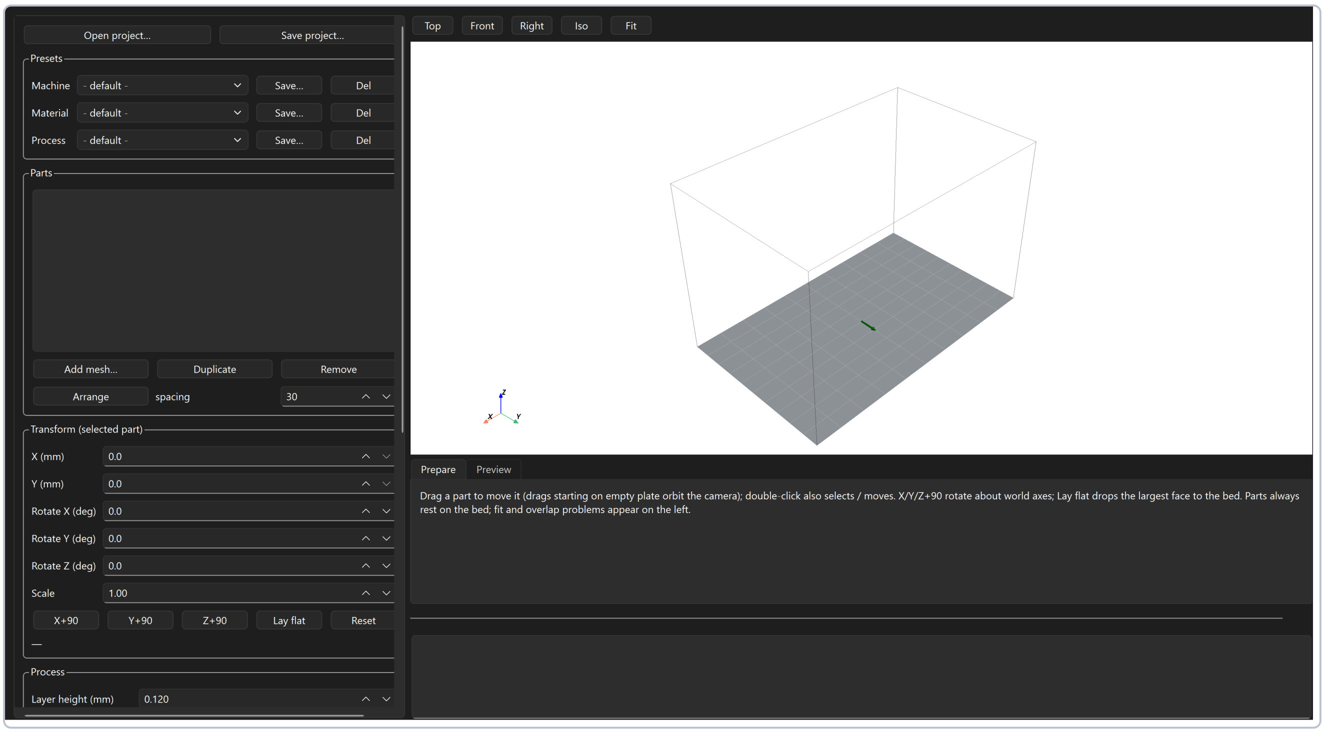
2. Starting the studio

- **One-file executable:** double-click `RotoforgeSlicer.exe` (or the *Rotoforge Studio* desktop shortcut). The first ~30 seconds are silent while the bundle unpacks — do not launch it twice. Windows SmartScreen may warn on an unsigned build: choose *More info → Run anyway*.
- **From a source checkout:** `python -m rotoforge_slicer.studio`.



The window has three regions:

Region	Contents
Left panel (scrollable)	Project open/save, Presets, Parts list, Transform, Process settings, Advanced parameters, process-window and CSV buttons, Slice / Save G-code
Viewport (top right)	The build plate with camera presets (Top / Front / Right / Iso / Fit); in Preview mode, the toolpath and the vertical layer-range slider
Tabs + log (bottom right)	Prepare / Preview tabs, the playback controls, and the log — read the log; every warning the slicer raises appears here

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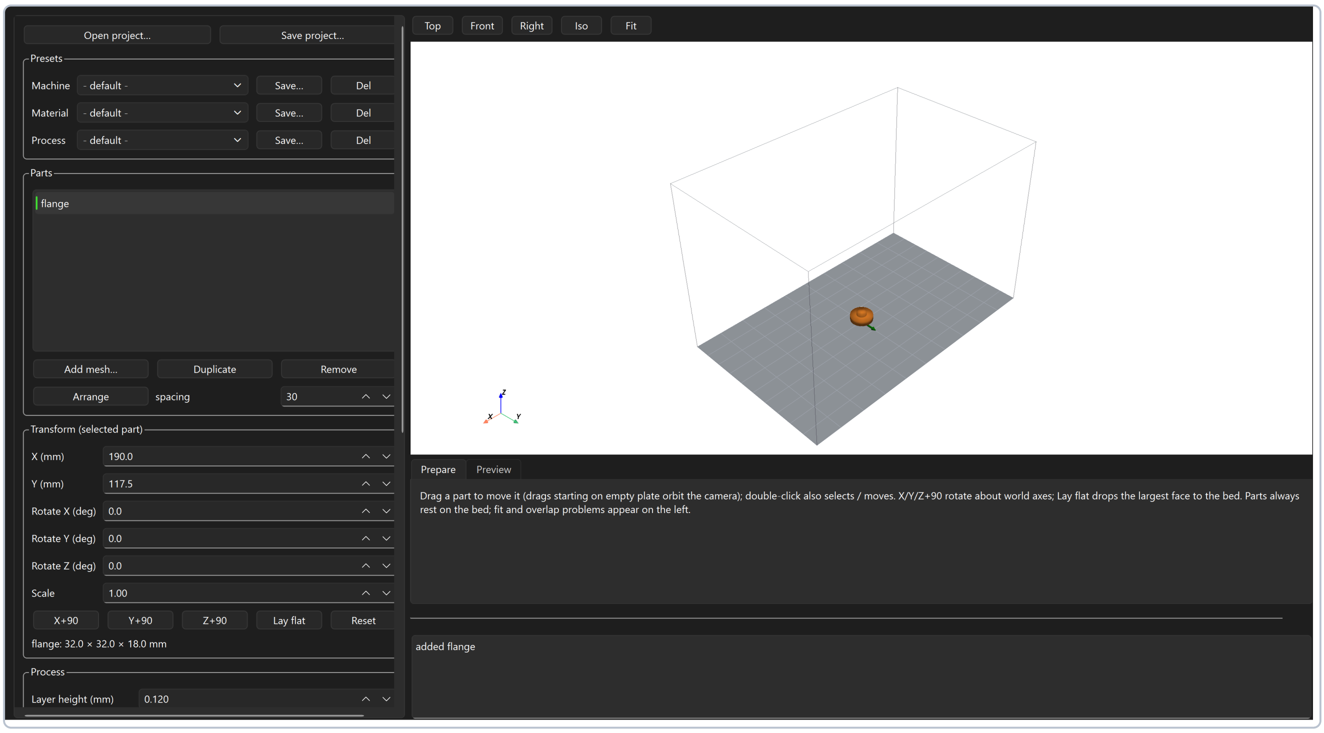
3. Step 1 — Import and place a mesh

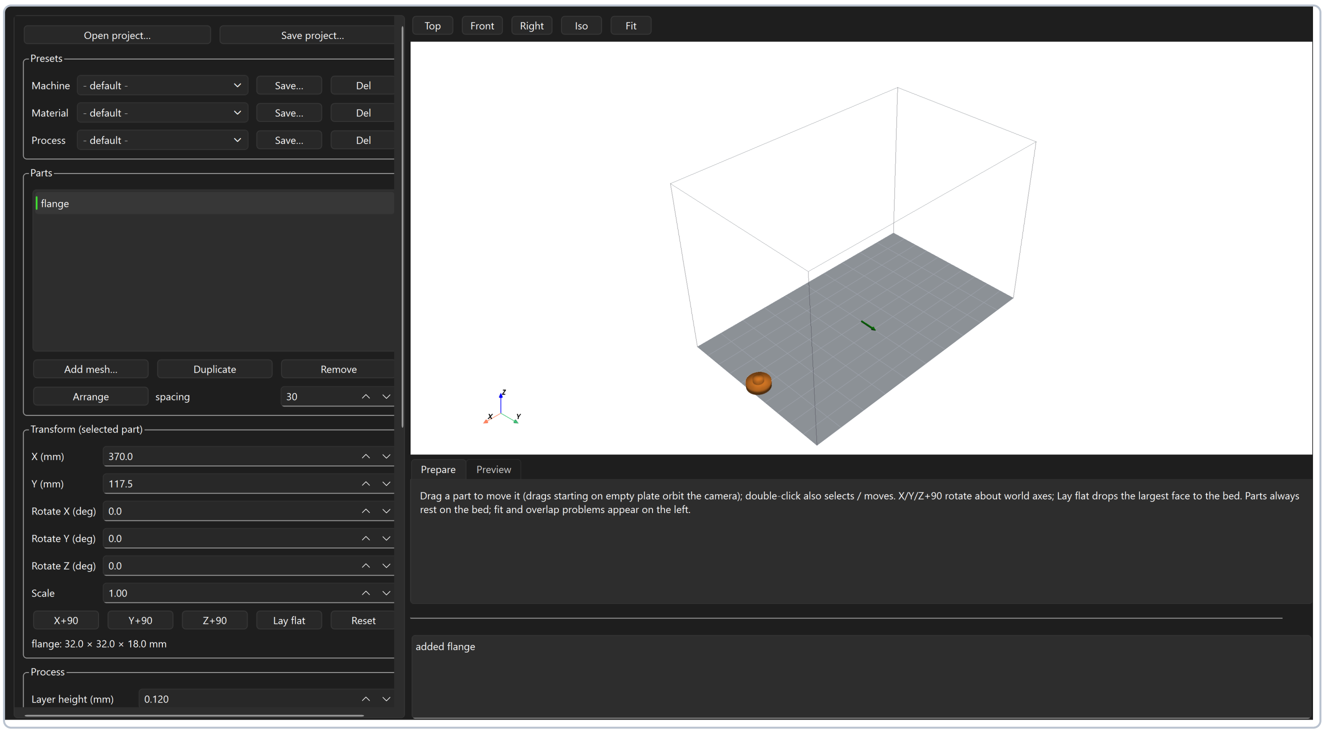
Click **Add mesh...** and pick an `.stl`, `.3mf`, `.obj`, or `.ply` file. The part lands at the plate centre, automatically dropped so it rests exactly on the bed — Z placement is never yours to manage.



Placing and orienting:

- **Select** a part in the Parts list, or double-click it in the viewport.
- **Move** it by dragging in the viewport (drags that start on empty plate orbit the camera instead), by double-clicking a target spot, or by typing X/Y in the Transform panel.
- **Rotate/scale** with the Rotate X/Y/Z and Scale fields (rotations tumble the part about its own middle), or use the quick buttons: **X+90 / Y+90 / Z+90** turn about the world axes; **Lay flat** drops the part's largest flat face onto the bed; **Reset** clears the orientation.
- **Duplicate / Remove** manage copies; **Arrange** auto-places everything with the given spacing — the default 30 mm keeps neighbouring parts clear of the 50 mm wheel body while it works.
- The dimensions readout under the buttons shows the placed bounding box.

Watch the red text. Placement problems appear immediately in the left panel — a part too close to the plate edge (every pass needs a lead-out runout, so an envelope is reserved on all four sides), taller than the build volume, or overlapping another part:



Slicing does not block on these warnings, but the emitted G-code will fail validation if a move would leave the machine volume — fix placement first.

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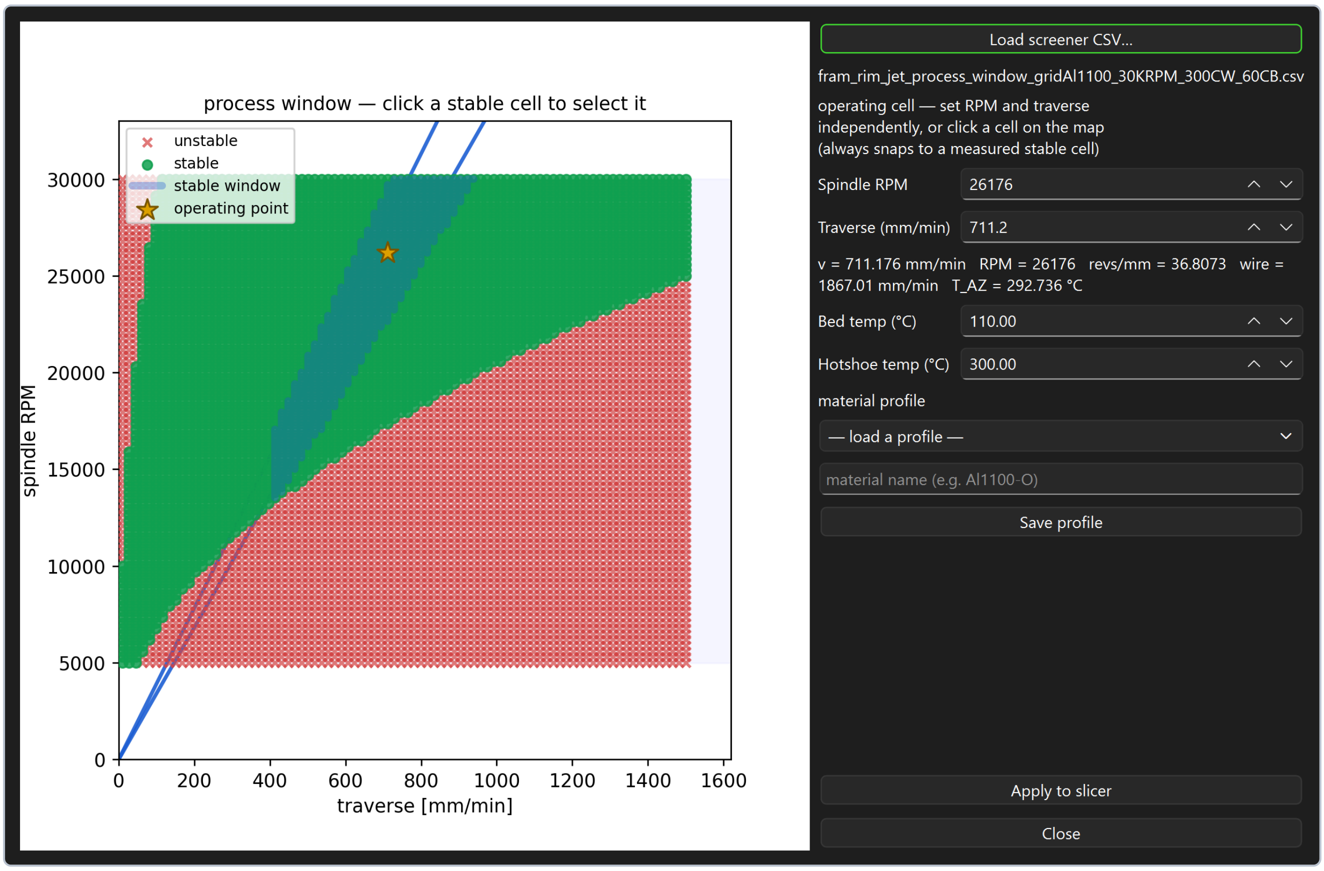
4. Step 2 — Load the process window (the material)

This is the most important step, and the one with no filament-slicer equivalent. The FRAM parameter screener characterizes your material and produces a CSV grid of measured cells: each cell is one (RPM, traverse) pair that was actually run, marked **stable** (deposits correctly) or **unstable** (cold, overheated, or otherwise bad). One stable cell fully determines a pass's RPM, traverse speed, and wire feed. The slicer **never interpolates** between cells — every number it uses was measured on hardware.

Open **Process window / material...** and load your screener CSV:



- The map plots every tested cell on the traverse × RPM plane: **green = stable**, **red = unstable**. The shaded horizontal band is the spindle's usable RPM window.
- **Spindle RPM** and **Traverse** are independent targets — type either and the selection snaps to the nearest measured *stable* cell. Or simply **click a cell on the map**. The gold star is your operating point; the highlighted band is the contiguous stable stretch of the constant-revs/mm ray your cell sits on (an unstable cell breaks that band — the slicer will not plan across it).
- The readout under the spins shows exactly what the cell means: traverse, RPM, the implied revs/mm, the wire feed rate, and the measured deposition zone temperature.
- **Bed temp** and **Hotshoe temp** select the matching tuned heater macros (`Hotshoe_300C.g` etc. — the macro must exist on the Duet).
- **Material profiles** save the whole selection (CSV + cell + temps) under a name for one-click reload later.
- **Apply to slicer** commits the selection. What the dialog displays is exactly what the slicer runs — if you don't Apply, nothing changes.

If you slice without any CSV, the studio uses a single-speed fallback (`emit.feed_dep_mm_min`) and says so — fine for motion tests, wrong for real deposition.

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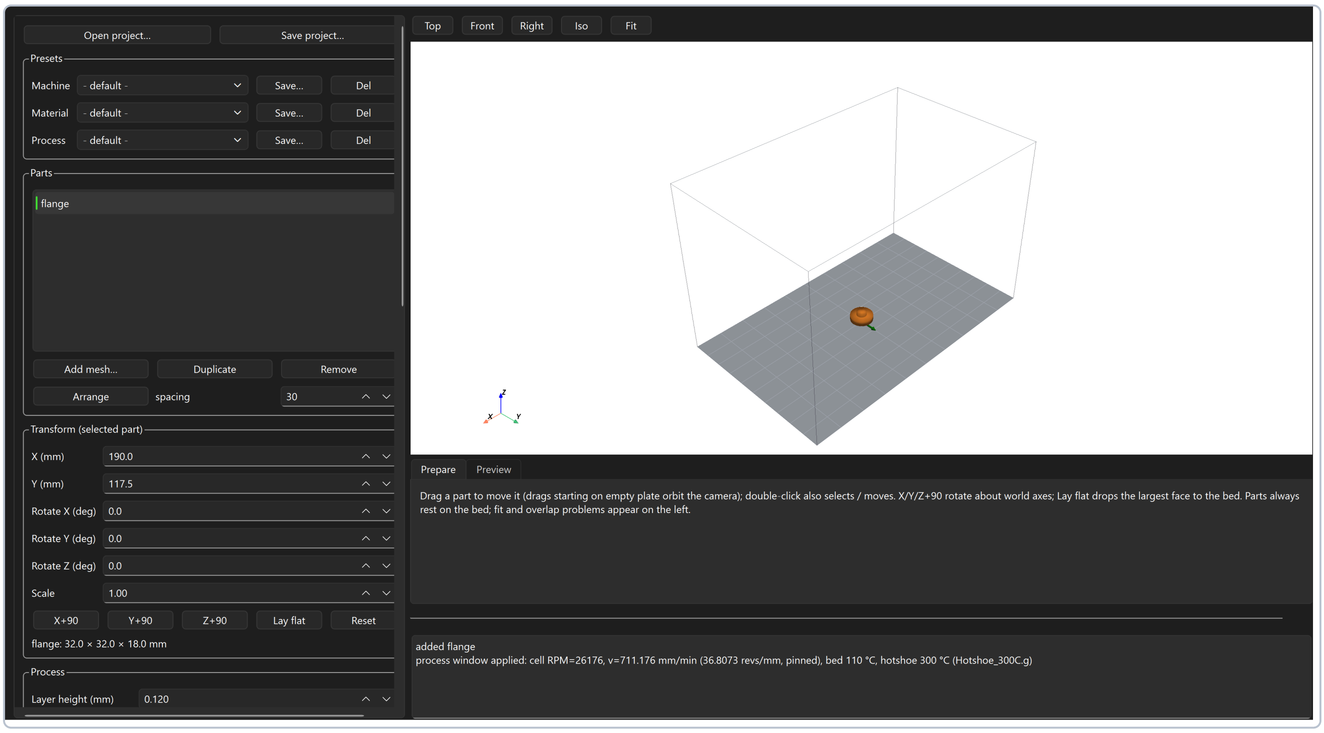
5. Step 3 — Pick the settings

The Process group

Setting	What it does
Layer height (mm)	Slice thickness (default 0.12). Must match what the process actually deposits per pass — this is a process property, not a quality knob.
Bead width (mm)	The deposited track width (default 1.0, the wheel rim). Sets wall offsets and raster pitch.
Raster overlap	Fraction of bead width by which neighbouring raster lines overlap (default 0.15). More overlap = denser fusion, more material.
Min deposit len (mm)	Hardware floor (default 6): passes shorter than this cannot form a stable deposit and are dropped. Features thinner than this simply do not fill — see Troubleshooting.
C-axis A min / A max (deg)	The wheel axis's usable continuous angular range, from machine calibration. The ±180° default is a placeholder; the real measured range decides whether closed loops can run as one pass and how freely ring seams can move.
Fill mode	`raster` — straight back-and-forth hatch, direction auto-chosen per region; `streamline` — boundary-following curved fill (fewer, longer passes on curvy parts); `contour` — concentric rings all the way in; `outline` — the outermost ring only.
Perimeter loops	With raster/streamline: lay N wall loops around the infill (0 = off). Walls deposit after the infill.
Ring seam	Where each closed ring starts: `extreme` (default — the winding-feasible start), `nearest` / `aligned` / `random` policies. At a 360°-wide axis range one-pass rings are pinned by physics — see the note in Troubleshooting.
Crosshatch	Alternate the fill heading between layers by ± the crosshatch angle.

Advanced parameters

Open the collapsible **Advanced parameters** group for the tuning fields:



Field	What it roughly does
Lead-in (mm)	Forward distance of the moving plunge that starts each pass — the wheel lands while already travelling.
Lead-out (mm)	Runout past the end of each pass where the wire is cut; reserved around the plate edge for every part.
Approach clearance (mm)	Airborne gap above the surface before the plunge begins.
Inter-pass lift (mm)	How high the head lifts between passes (≥ 10 mm keeps the disc clear of built material).
Wire diameter (mm)	The feedstock diameter — used by extrusion accounting.
Travel / Z / Deposition-fallback feeds (mm/min)	Airborne XY speed, vertical speed, and the deposition speed used only when no screener CSV is loaded.
C-axis slew ω_C (deg/s)	How fast the wheel axis can turn. Sets the minimum curve radius at a given traverse (R ≥ v/ω) — curves tighter than this split into separate passes.
Corner scrub budget (deg·mm)	How much heading step × following-segment length is tolerated at a vertex before the corner is split into an airborne reorient. Lower = stricter corners.
Collision clearance (mm)	Required gap between the 50 mm wheel body and already-deposited material.
Collision wire lead (mm)	How far the fragile leading wire is assumed to reach ahead of the contact point.
Crosshatch angle (deg)	The ± heading offset used when Crosshatch is on.
Streamline step / curl	Streamline integration step and how strongly the fill follows the boundary (0 = straight).
Contour simplify (mm)	Ring simplification tolerance — keeps G-code size sane without visibly changing the path.

Field	What it roughly does
Seam align radius (mm)	How far the `aligned` seam policy searches for the previous layer's seam.
prefer one-pass rings	Keep ring seams inside the winding seat window so closed rings deposit in one pass. Off: seams move freely at the cost of one extra airborne unwind + wire cut per ring.
dry run	Emit motion-only G-code: no spindle, no heaters, no wire. **Use this for the first run of any new job on hardware.**

Presets and projects

The three preset selectors at the top — **Machine / Material / Process** — save and recall named bundles of these settings (`Save…` writes the current values; a `(modified)` suffix means your current settings differ from the saved preset). **Save project...** writes everything — the plate with embedded meshes, all settings, and the screener CSV — into a single `.rfproj` file that **Open project...** restores exactly; it is also the perfect bug-report attachment.

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6. Step 4 — Slice

Press **Slice**. The pipeline runs off the UI thread: slicing the mesh into layers, choosing the operating point, planning passes under all the machine rules, checking collisions, and emitting + validating G-code. The progress bar tracks the stages; a big part with a dense screener grid takes a few extra seconds in the "selecting operating point" stage.

When it finishes, read the summary in the log:

```
passes: 4758
operating point: RPM 22647, traverse 711.176 mm/min, revs/mm 31.844
operating point: revs/mm=36.8 ray, v=[413,939] mm/min, nominal v=711 -> RPM=22647, ...
collisions: none (SPEC §4.6)
validation: all §6.3 checks pass; G-code ready to save
```

- **passes** — how many separate deposition passes the plan contains.
- **operating point** — the screener cell actually used (RPM, traverse, wire feed, measured temperature).
- **collisions** — result of sweeping the 50 mm wheel disc and the leading wire against already-built material.
- **validation** — the emitter's §6.3 proof of every hardware rule. Only a clean pass enables **Save G-code...**.
- **NOTE:** lines report non-fatal planner decisions (for example a seam policy constrained by the axis range).

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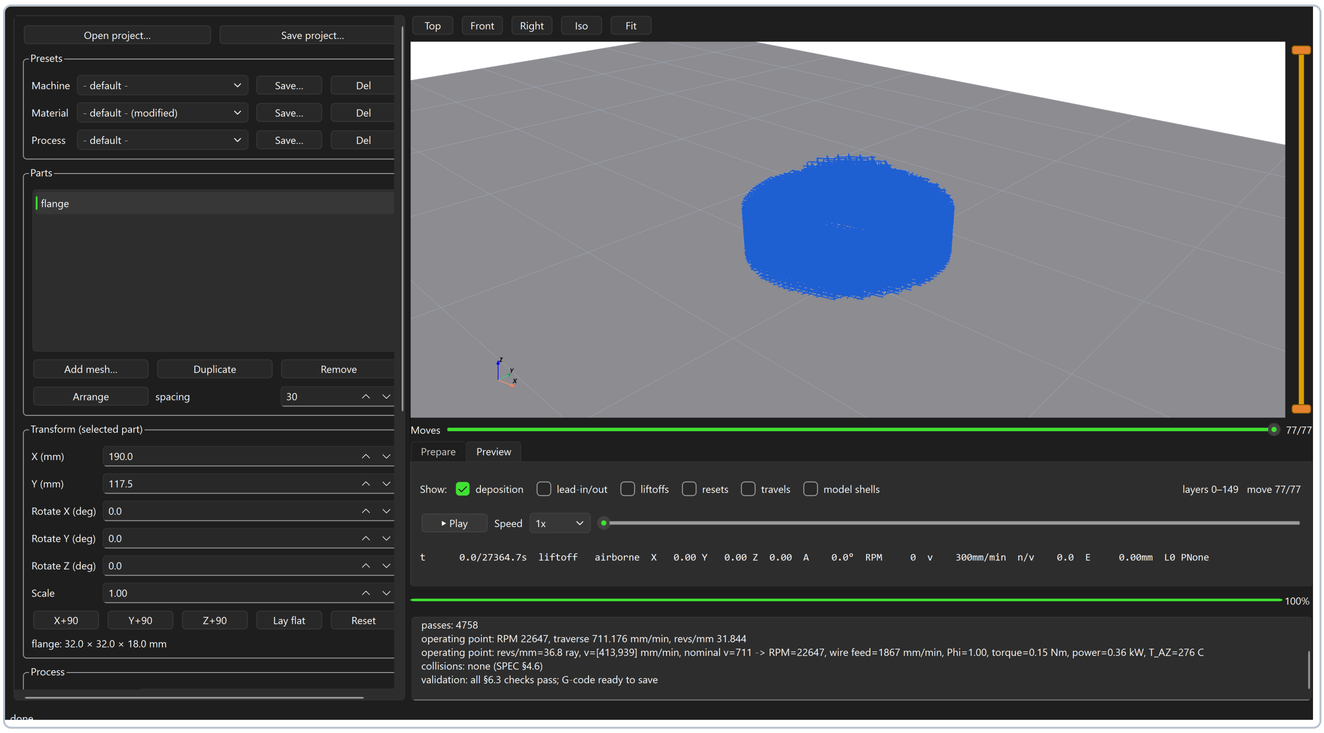
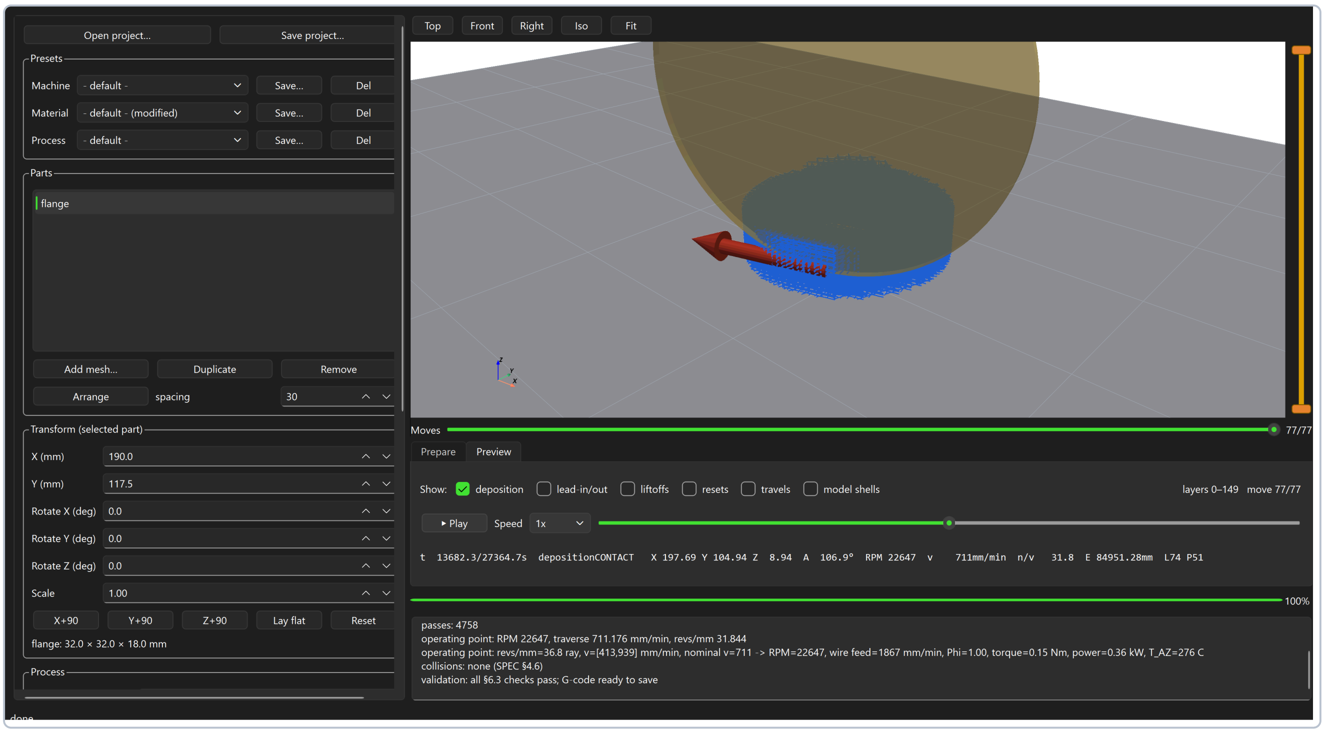
7. Step 5 — Preview and simulate

Slicing switches to the **Preview** tab: the mesh is hidden and the actual toolpath is shown (tick **model shells** to ghost the mesh back in).



- **Move-class toggles** — deposition is shown by default; add lead-in/out, liftoffs, resets, and travels to see the airborne structure.
- **The vertical slider** (right edge of the viewport) is the layer-range window — drag either handle to isolate layers.
- **The Moves slider** (under the viewport) reveals the top visible layer move by move — step through exactly what the machine will do.

Play runs a kinematic simulation with emitter-faithful timing — the moving head is drawn as the real vertical 50 mm wheel with its heading arrow tracking the commanded A axis, including plunges, dwells, and unwinds:



The readout shows time, contact state, position, wheel angle A, RPM, traverse, revs/mm, and cumulative wire E. Scrub the timeline to any instant.

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8. Step 6 — Save the G-code

Save G-code... writes the validated RRF file. If collisions were detected you are warned first — the CLI would refuse outright; saving anyway is on you. On the machine: run a **dry run** file first for any new geometry, keep the first live run attended, and remember the G-code was emitted for the machine calibration in `config/machine_duet3.yaml` — if your axis range, steps, or slew were recalibrated since, re-slice.

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9. Troubleshooting

Symptom	Cause	Fix
Red text: "*outside the 380x235 plate (footprint ... with the N mm lead-out envelope)*"	Every pass ends in a lead-out runout, so parts must keep that margin from every plate edge	Move the part inward, or press **Arrange**
Red text: "*footprints overlap*"	Two parts' XY footprints intersect	Separate them; keep ≥ wheel-body spacing (Arrange does)
Log: "*no CSV (single-speed fallback)*"	Slicing without a process window	Load the screener CSV (Step 2) unless this is a motion test

Symptom	Cause	Fix
Label: "*CSV MISSING: path*"	A preset/profile/project references a screener CSV that is not on this machine	Re-pick the CSV in the process window; projects carry an embedded copy automatically
Error: "*traverse_target … lies outside the ray's contiguous stable run*"	A saved profile's operating point does not exist in the currently loaded CSV (stale profile / different export)	Open the process window and re-pick the cell; re-save the profile
VALIDATION FAILED (§6.3)	The emitter proved a hardware rule would be violated — e.g. RPM outside the spindle window, a heading step too sharp for in-contact travel, axis range exceeded	This is the safety net. Read the message: pick a different screener cell (RPM window), let corners split (scrub budget), or fix the A range. Never try to bypass it
COLLISIONS: n (see red rings)	The swept wheel disc or leading wire would hit already-deposited material	Inspect the red rings in Preview; reorient the part, reduce height near the conflict, or increase collision clearance; Save warns before letting you keep a colliding file
NOTE: "*seam policy … constrained to a N-vertex seat window*"	Physics: at a 360°-wide axis range, a one-pass ring's seam is pinned where the axis meets its range stop	Calibrate and enter the real (wider) axis range, or untick **prefer one-pass rings** — that scatters seams at the cost of one extra unwind + wire cut per ring
Thin ribs/features come out empty	Features needing passes shorter than **Min deposit len** (6 mm) or thinner than one bead get no stable deposit	Thicken the feature (≥ bead width; lengths ≥ 6 mm), or accept the void — the limit is physical, not a setting to lower casually
Some layers have far fewer passes than expected	Regions shrank below the bead/min-length floors at that height	Check those layers in Preview with the layer-range slider
Slice feels slow on "selecting operating point…"	Dense real screener grids (~7 400 cells) take a few seconds to search	Normal; the result is cached per slice
Nothing appears after slicing / empty plan	Mesh not watertight (repair is best-effort), wrong units (a 0.03 mm "part"), or the part is entirely below minimum feature sizes	Check the dimensions readout after import; repair/rescale the mesh
Exe does nothing for ~30 s / SmartScreen blocks it	One-file bundle unpack; unsigned binary	Wait it out (don't double-launch); *More info → Run anyway*
Startup log shows "*presets: … bad value … kept base*"	A hand-edited preset file contains an invalid value	The studio kept the safe base value; fix or delete the file in `config/presets/` (`~\.rotoforge\presets\` for the exe)
Preset shows **(modified)**	Your current settings differ from the saved preset	`Save…` to update the preset, or re-select it to discard the edits

Where things live: machine calibration in `config/machine_duet3.yaml` (the source of truth — recalibrations belong there); presets and studio state in `config/presets/` and `config/studio_state.yaml` (source checkout) or `~\.rotoforge\` (frozen exe); the full specification in `docs/rotoforge_slicer_SPEC.md`; design decisions in `docs/DECISIONS.md`. There is also a command-line interface: `rotoforge-slice --help`.